RUBRIC COMPONENT #2 (3 MARKS)	OFFICIAL CRITERIA: Relational schema with keys, relationships, and constraints

Relational Schema, Keys, Integrity & Normalization Proofs

Course: DBMS Mini Project (UE24CS395) • **Institution:** PES University (2026–2027)
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1. Primary Key, Foreign Key & Domain Constraints Matrix

Every table in the schema enforces strict domain constraints, primary keys, and foreign keys. Associations use ON DELETE CASCADE while legal custody uses ON DELETE RESTRICT.

Constraint Category	Target Table & Columns	Enforcement Rule & Purpose
PRIMARY KEY (Composite)	12 Junction Tables (e.g. case_people)	PRIMARY KEY (case_id, person_id) ensures uniqueness on paired association
UNIQUE Hardware Key	vehicles(registration_number), devices(serial_number)	Prevents duplicate hardware exhibits or counterfeit license registrations
UNIQUE Personnel Key	officers(badge_number, contact_email)	Guarantees one cryptographic account per active law enforcement badge
CHECK (Status Domain)	cases(status)	CHECK (status IN ('Open', 'Under Investigation', 'On Hold', 'Closed'))
CHECK (Priority Domain)	cases(priority)	CHECK (priority IN ('Low', 'Medium', 'High', 'Critical'))
CHECK (RBAC Role)	officers(role)	CHECK (role IN ('ADMIN', 'INVESTIGATOR')) strictly bounds auth roles
FK CASCADE	case_people, evidence, case_activity	ON DELETE CASCADE purges dependent junction records when parent is removed
FK RESTRICT (Custody)	evidence_custody(officer_id)	ON DELETE RESTRICT blocks deleting officers with active custodial responsibility

2. Normalization Analysis & Formal Proofs (1NF, 2NF, 3NF, BCNF)

First Normal Form (1NF): All column values are atomic scalar values. No repeating groups or multivalued delimited strings. Multiple phone numbers or hardware items are decoupled into person_devices.
Second Normal Form (2NF): Schema is in 1NF and contains zero partial key dependencies. In composite junction tables like case_people(case_id, person_id), descriptive attributes (role_in_case, added_at) depend fully on the complete composite key.
Third Normal Form (3NF): Schema is in 2NF and contains zero transitive dependencies. Functional dependency proof: In an unnormalized relation, case_id -> lead_officer_id and lead_officer_id -> department. Decomposing into cases and officers eliminates this transitive anomaly.
Boyce-Codd Normal Form (BCNF): For every non-trivial functional dependency X -> Y, X is a superkey.